

Scholar: System & Architecture Master Guide

Smart Companion for Holistic Organization, Literature Analysis & Research

Local-First, Privacy-Preserving Scientific RAG with Verifiable Page Citations & Multimodal Grounding

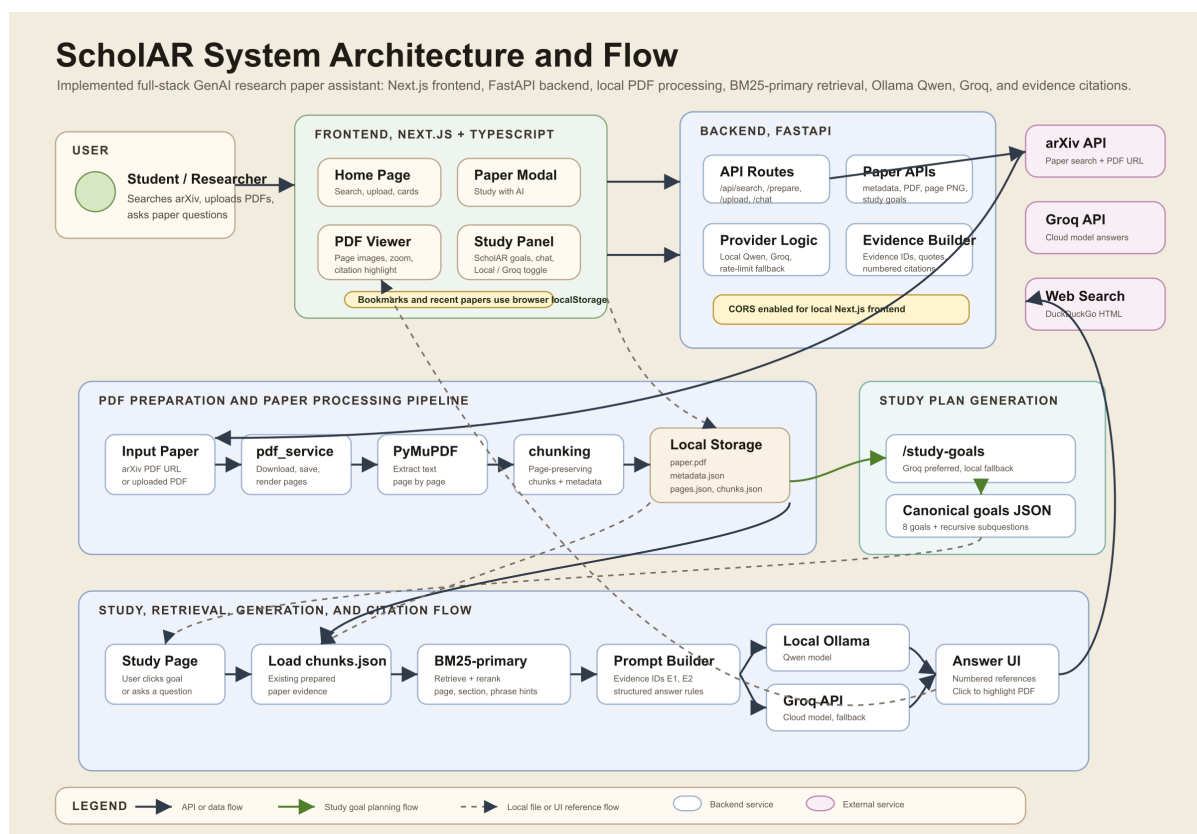
Executive Summary

Scholar is an on-device AI research assistant that operates 100% locally via Ollama:

- **Zero Cloud Leakage:** Ingests arXiv papers or custom PDFs; inference, vision processing, and embeddings remain entirely on-device.
- **Verifiable Citations:** Maps short-lived evidence tokens ([E1], [E2]) to immutable PDF pages, eliminating hallucinated page numbers.
- **Visual Grounding:** Directly routes figure and table queries to a local Vision LLM with high-resolution 3× zoom image crops.
- **Multi-Document Reasoning:** Resolves bibliographies via Semantic Scholar API to dynamically ingest and cross-read referenced works.

1 System Architecture & Data Flow

Scholar decouples PDF parsing, lexical/dense retrieval, local model routing, and evidence-ID citation translation into a modular, high-throughput pipeline.



1.1 Core Services Map

Component	Source File	Core Functionality
API Entrypoint	backend/main.py	FastAPI orchestration, REST endpoints (<code>/chat</code> , <code>/study-goals</code> , <code>/references</code>).
PDF Extraction	backend/services/pdf_service.py	PyMuPDF text parsing, bounding-box detection, 3× zoom page PNG rendering.
Chunking	backend/services/chunking_service.py	Page-preserving sliding window (1400 words, 120 overlap). Never crosses pages.
Retrieval	backend/services/retrieval_service.py	BM25 primary scoring + page hint (+1.25), section (+0.5), and visual (+1.5) boosts.
Ollama Router	backend/services/ollama_service.py	Ollama client managing context windows for <code>qwen3.5:9b</code> , <code>gemma4:12b</code> , etc.
Vision Service	backend/services/vision_service.py	Multimodal figure Q&A with base64 image encoding and caption fallback.
Reference Service	backend/services/reference_service.py	Semantic Scholar bibliography resolution and S2 title-search for uploaded PDFs.
Frontend UI	frontend/app/ & components/	Next.js 15 + React workspace (<code>StudyWorkspace</code> , <code>PdfViewer</code> , <code>ChatBox</code>).

2 Deep-Dive into RAG Pipeline Components

2.1 1. Ingestion & Page-Preserving Chunking Strategy

Standard RAG frameworks split text by arbitrary token lengths, frequently cutting across page boundaries. When an LLM cites that chunk, it cannot point to an exact single page.

- **Our Rule:** One Chunk ∈ One Page. No chunk ever spans across multiple pages.
- **Sliding Window Parameters:** Target word count = 1400 words, word overlap = 120 words per page.
- **Visual Crop Extraction:** Bounding box detection automatically crops figures/tables at 3× **zoom** for optical clarity.

2.2 2. Embedding Model & Primary Retrieval Engine

- **Lexical Primary (BM25):** Used as the primary signal because scientific literature requires exact matching on numbers, dataset names, formulas, and parameters (e.g., *BLEU 28.4*, *lr=1e-4*, *WMT14*).
- **Dense Embedding Model:** Pure PyTorch implementation of `all-MiniLM-L6-v2` (384-dimensional dense vectors). Runs completely locally without heavy external dependencies.
- **Reciprocal Rank Fusion (RRF, $k = 60$):** Combines lexical and dense ranks for research evaluation:

$$\text{RRF Score}(d) = \frac{1}{60 + \text{Rank}_{\text{BM25}}(d)} + \frac{1}{60 + \text{Rank}_{\text{Dense}}(d)}$$

2.3 3. Heuristic Re-Ranking Boosts

Before passing chunks to the LLM, BM25 scores are dynamically adjusted based on query intent:

- **Page Hint Boost (+1.25):** If the query asks *"What is discussed on page 4?"*, page 4 chunks are prioritized.
- **Section Hint Boost (+0.5 to +1.0):** Boosts Abstract, Methods, Results, or Limitations sections when implied.
- **Visual Cue Boost (+1.5):** If a query mentions *"Figure 2"* or *"Table 1"*, figure chunks jump to Rank 1.
- **Query Expansion:** Synonym expansion for research vocabulary (e.g., *result* → *accuracy*, *BLEU*, *F1*, *score*).

2.4 4. Multimodal Vision Routing

When a query targets a visual element:

1. Retrieval ranks the figure chunk as top-1.
2. The high-resolution 3× zoom PNG crop of that figure is base64-encoded.
3. Sent directly to the local Vision LLM (`qwen3.5:9b` or `gemma4:12b`) along with the question.
4. **Caption Fallback:** If the vision model is unreachable, it automatically falls back to caption text QA.

2.5 5. Indirect Evidence ID Citation Grounding

To eliminate invented or hallucinated page numbers:

1. **Prompt Ingestion:** Retrieved chunks are labeled as [E1], [E2], [E3] in the LLM prompt.
2. **Constrained Output:** The model is strictly instructed to cite using only [E1], [E2] and cannot write page numbers.
3. **Application Translation:** Backend maps [E1] to the source chunk's verified metadata → rendering a clickable [1] (Page 3) chip in the UI.
4. **Guarantee:** 100% of cited page numbers exist on real PDF pages by construction.

2.6 6. Multi-Document Expansion (Citation Graph)

- **Semantic Scholar API:** Queries S2 by arXiv ID or paper title to fetch up to 500 reference papers.
- **On-Demand Ingestion:** Users can ingest any cited paper with one click. It is downloaded, chunked, and tagged with `source_paper_id`.
- **Cross-Paper Retrieval:** Queries search across the anchor paper and all ingested secondary papers simultaneously.

3 Academic Baseline Papers & Related Works

Baseline Paper	Focus Area	Comparison to Scholar
OpenScholar (Asai et al., 2024)	Cloud Scientific RAG	Relies on a 45M+ paper cloud datastore and frontier LLMs. Scholar is 100% on-device and privacy-preserving.
PaperQA2 (Skarlinski et al., 2024)	Scientific Agents	Uses Rerank-Clean-Summarize (RCS) agent loop. Reimplemented locally as our primary baseline.
SciRAG (Ding et al., 2025)	Outline-Guided RAG	Citation-aware multi-doc synthesis; informed our human eval precision/recall framework.
M3SciQA (Li et al., EMNLP 2024)	Multimodal Scientific QA	Benchmark for finding which reference answers a figure question. Scholar scored 0.474 MRR .
ColPali (Faysse et al., ICLR 2025)	Visual Doc Retrieval	Patch-level VLM page embeddings; theoretical foundation for our planned visual retrieval engine.
SummaC (Laban et al., 2022)	NLI Faithfulness	Entailment consistency metric; forms Tier 1 of our NLI-CFS scorer.
FActScore (Min et al., 2023)	Atomic Fact Verification	Atomic claim decomposition; foundation for our factual precision evaluations.
BEIR (Thakur et al., 2021)	Zero-Shot Retrieval	Established BM25 superiority over dense embeddings on specialized technical domains.
SlideTailor (Zeng et al., 2026)	PDF Slide Generation	Upstream PDF extraction bottleneck mirrors Scholar’s extraction goals.

4 What Is Done (Completed Inventory)

- ☑ **Full Backend & REST API:** Complete ingestion, chunking, retrieval, vision, and reference routing.
- ☑ **Next.js 15 Frontend:** Resizable split layout, PDF viewer, interactive citation chips ([1]), inline figure cards.
- ☑ **100-Case / 25-Paper Benchmark:** Mined and source-verified covering 8 scientific claim types.
- ☑ **NLI-CFS & LLM Entailment Judge:** 3-tier faithfulness metric + strict LLM contradiction detector.
- ☑ **Visual Grounding Pipeline:** 3× zoom page crop rendering with local VLM reasoning (qwen3.5:9b).
- ☑ **Multi-Document Ingestion:** S2 API bibliography resolution with individual reference paper ingestion.
- ☑ **Research Paper Manuscript:** 8-page compiled draft (paper/manuscript.pdf) and compliance checklist.
- ☑ **Human Evaluation Interface:** Offline blinded scoring tool (score_sheet.html) with 350 pre-generated model outputs.

5 Experimentation Results & Key Benchmarks

5.1 A. Primary Retrieval Benchmark (100 cases, 25 papers)

Finding: Lexical BM25 significantly outperforms Dense MiniLM embeddings on technical text.

Retrieval Method	Recall@1	Recall@5	MRR
dense_only (all-MiniLM-L6-v2)	0.470	0.740	0.572
keyword_overlap	0.670	0.950	0.779
bm25_primary (Ours)	0.810	0.930	0.861

5.2 B. Resource-Matched Local Baseline Comparison (91 cases on qwen3.5:9b)

Finding: Scholar leads in generation correctness and faithfulness while guaranteeing valid page citations.

System	Generation Faithfulness	Answer Correctness	Citation F1	Provenance Valid
Long-Context PDF-Chat	0.801	0.267	0.742	No
Vanilla RAG	0.860	0.340	0.802	No
PaperQA2-Style (Local RCS)	0.872	0.550	0.782	No
Scholar (Ours)	0.892	0.563	0.760	100%

5.3 C. M3SciQA Multi-Document Locality Benchmark (297 cases)

Finding: Visual entity resolution unlocks cross-document search where text retrieval fails.

System	Deployment Mode	MRR	Recall@5
Human Expert (<i>published</i>)	Manual Review	0.796	—
GPT-4o (<i>published</i>)	Cloud Frontier API	0.500	—
Scholar + Local Vision (qwen3.5:9b)	Local Laptop (On-Device)	0.474	0.606
Scholar + Local Vision (gemma4:12b)	Local Laptop (On-Device)	0.455	0.572
GPT-4V (<i>published</i>)	Cloud Frontier API	0.400	—
text-embedding-3-large (<i>published</i>)	Cloud API	0.297	—
Scholar Text-only BM25	Local Laptop	0.180	0.242
Best Open-Source LMM (<i>published</i>)	Cloud / Server	0.144	—

5.4 D. Abstention on Unanswerable Queries (20 negative control cases)

Finding: Scholar abstains rather than hallucinating when information is absent.

- **qwen3.5:9b: 100% Abstain** (0% hallucination / fabrication)
- **llama3.1:8b: 100% Abstain** (0% hallucination / fabrication)
- **gemma4:12b: 95% Abstain** (5% partial overlap)
- **mistral:7b: 90% Abstain** (10% partial overlap)

5.5 E. On-Device Hardware Efficiency (Apple Silicon 18 GB RAM)

- **BM25 Retrieval Latency:** ~40 ms (instantaneous).
- **Generation Throughput:** 22–27 tokens/s across all 7B–12B local models.
- **End-to-End Turnaround:** 6.1 s (mistral:7b), 6.3 s (llama3.1:8b), 11.4 s (qwen3.5:9b), 8.8 s (gemma4:12b).

6 Critical Research Takeaways (Honest Findings)

1. **Cosine Faithfulness Inflates Scores by 2×:** Embedding cosine similarity cannot catch contradictions (“Model A works” vs “Model A fails” look identical in vector space). Our strict LLM Entailment Judge fixes this and provides honest grounding metrics (~0.61 vs inflated ~0.85).
2. **Provenance vs. Faithfulness:** Ephemeral evidence IDs guarantee that 100% of cited page numbers exist, but claim support still depends on retrieval quality (66% support rate). ID grounding buys provenance and auditability.
3. **The M3SciQA Win:** Breaking figure questions into two steps (Vision VLM reads figure → BM25 ranks bibliography) delivers **0.474 MRR** locally—beating open-source baselines and approaching GPT-4o.

7 Future Roadmap & Unfinished Backlog

- **Human Study Execution:** Run the 350-case human evaluation on `score_sheet.html` and compute Krippendorff’s α .
- **Structured Document Parsing:** Benchmark Nougat / Grobid against PyMuPDF for table and equation preservation.
- **ColPali Visual Page Retrieval:** Index full PDF pages directly with VLM patch embeddings.

- **LaTeX Math Tokenizer & UI Rendering:** Add symbol-level math tokens and KaTeX rendering to the chat interface.
- **Interactive Citation Graph:** Build a Connected Papers-style D3 force-directed citation network in the UI.

8 Developer Quickstart & Runbook

8.1 Prerequisites & Setup

```
# 1. Start Ollama and pull default models
ollama serve
ollama pull qwen3.5:9b      # Default model
ollama pull gemma4:12b     # Optional evaluation model

# 2. Start the Backend API (FastAPI on http://localhost:8000)
make backend

# 3. Start the Frontend UI (Next.js on http://localhost:3000)
make frontend
```

8.2 Evaluation Execution

```
python3 evaluation/run_retrieval_eval.py --scaled    # Scaled 100-case retrieval
python3 evaluation/run_comparison_eval.py           # Baseline comparisons
python3 evaluation/run_abstention_eval.py           # Abstention evaluation
python3 evaluation/run_efficiency_eval.py           # Latency & memory profile
```